Data Science Project Report: Titanic Data Analysis

Course: Data Science and Artificial Intelligence

Institution: Philadelphia University, Jordan

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1. Introduction :

   The sinking of the RMS Titanic is one of the most infamous maritime disasters in history. On April 15, 1912, during her maiden voyage, the Titanic sank after colliding with an iceberg. While there was some element of luck involved in surviving the disaster, it clear that some groups of people were more likely to survive than others, such as women, children, and the upper-class.
2. The Core Problem Statement:

   In the domain of Data Science, the Titanic dataset serves as a benchmark study for understanding how structured socio-demographic configurations dictate structural patterns of human survival. The core challenge addressed in this investigation is to systematically dissect the dataset to prove that survival was not a purely stochastic (random) phenomenon. Instead, survival rates were deeply governed by systemic hierarchies, socio-economic positioning, gender-based cultural protocols, and familial structural networks.

3. The Goal Statement:

The objective of this project is to perform a comprehensive Exploratory Data Analysis (EDA) and Data Preprocessing on the Titanic passenger dataset. The goal is to analyze the key factors that influenced passenger survival rates (such as gender, age, socio-economic status, and family size), and clean the data to make it structured for analytical insights.

2. Dataset Description

The dataset used in this project is the famous Titanic dataset obtained from **Kaggle**. It consists of a training sample containing information about passengers aboard the ship, repository (available at: https://www.kaggle.com/c/titanic/data).

Dataset Dimensions and Structure

- **Total Samples:** 891 rows (passengers).
- **Total Features:** 12 columns.
- **Target Variable:** `Survived` (0 = No, 1 = Yes).

Granular Attribute Decomposition

An exhaustive structural audit via `df.info()` classifies the 12 features into distinct statistical datatypes:

1. **`PassengerId` (Discrete Integer Index):** A continuous identifier ranging from 1 to 891 assigned arbitrarily to each record. It possesses zero statistical variance regarding the target output.
2. **`Survived` (Categorical, Binary Target):** The critical outcome parameter encoded as a binary flag:
    a. 0: Denotes non-survival (Deceased).
    b. 1: Denotes survival.
3. **`Pclass` (Ordinal Categorical Factor):** A metric reflecting socio-economic class positioning:
    a. 1: First Class (Upper economic tier).
    b. 2: Second Class (Middle economic tier).
    c. 3: Third Class (Lower economic tier / Steerage).
4. **`Name` (Unstructured Text / String):** The passenger's full name, which inherently contains embedded social honorifics and titles.
5. **Sex (Categorical Nominal String):** The biological gender designation of the individual (`male` or `female`).
6. **Age (Continuous Numeric):** Fractional or integer age tracking. Values lower than 1 represent infants.
7. **`SibSp` (Discrete Integer Count):** Quantifies the passenger's horizontal familial relations on board (number of siblings and/or spouses).
8. **`Parch` (Discrete Integer Count):** Quantifies the passenger's vertical familial relations on board (number of parents and/or children).
9. **`Ticket` (Alphanumeric String Identifier):** Raw structural string data tracking specific ticket registration values.
10. **`Fare` (Continuous Floating-Point Value):** The financial cost paid in British Pounds (£) for boarding passage.
11. **`Cabin` (Alphanumeric Structural String):** The specific spatial cabin placement marker indicating deck assignment and room allocation.
12. **`Embarked` (Nominal Categorical Character):** Specifies the physical port of boarding:
    a. S: Southampton, UK.
    b. C: Cherbourg, France.
    c. Q: Queenstown, Ireland.

Structural Gaps and Null Value Inventory

Executing an algorithmic missingness analysis using `df.isnull().sum()` revealed major missing entries across three distinct columns:

Total Observations = 891

- **`Cabin` Deficiency:** 687 missing observations (77.10\% missingness rate). The feature is severely fragmented, rendering direct numerical synthesis unfeasible.
- **Age Deficiency:** 177 missing observations (19.86\% missingness rate). Since age is a major biological indicator, simple deletion of rows would inject extreme survival bias into the data.
- **`Embarked` Deficiency:** 2 missing observations (0.22\%missingness rate). This is a minor localized gap suitable for simple statistical mode replacement.

Statistical Aggragation and Pivoting

To uncover initial relationships between demographic attributes and survival, we applied multi-axis data pivoting via Pandas `.groupby()` and `pd.crosstab()`. This phase deliberately isolated survival probability distributions against socio-economic status (`Pclass`), gender (`Sex`).

Advanced Visual EDA Architecture

- Visual validation was executed using the Matplotlib and Seaborn libraries to map complex interactions:
- **Target Balance Identification:** Constructing paired single-variable plots (pie charts and count-bars) to confirm the baseline distribution of the target variable (`Survived`).
- **Conditional Distribution Density Mapping:** Leveraging `sns.FacetGrid` to map continuous Age distributions directly against binary `Survived` outcomes across a 20-bin histogram mesh grid.
- **Multi-Layered Intersectional Evaluation:** Deploying a 3×2 matrix layout through `sns.FacetGrid` to map continuous Age behavior across the intersection of `Pclass` (stratified along vertical rows) and `Survived` outcomes (stratified along horizontal columns).
- **Categorical Point and Factor Configurations:** Building point-plots using `sns.catplot` to evaluate survival rates across `Pclass` variations, segmented explicitly by passenger groups and boarding locations.

Step 3: Data Cleaning & Imputation (Completing & Correcting)

To fix structural gaps discovered during the EDA phase:

1. The 'age' variable was refined by reducing the number of categories from 14 to 5 distinct groups. Following this, missing values were handled, and the mean was calculated to ensure data consistency for subsequent analysis.
2. **Dropping Features:**Dropped `Cabin` (due to >77% missing values)
3. Imputing Embarked: Replaced the 2 missing values with 'C', which was statistically identified as the dataset's most frequent port .

Cross-tabulation reveals an extreme survival split along gender lines:

Gender (Sex)	**Deceased Count**	**Survived Count**	**Survival Probability**
Female	81	233	**74.20%**
Male	468	109	**18.89%**

This 55.31\% survival difference proves the strict structural execution of the "women and children first" maritime evacuation protocol.

4.3 The Socio-Economic Privilege Vector (`Pclass`)

Socio-economic classification functioned as an institutional gatekeeper for survival access:

Survival Rate Pattern: {Pclass 1 } (62.96%) > {Pclass 2 } (47.28%) > Pclass 3 } (24.23%)

Passengers in First Class had a survival rate nearly three times higher than those trapped in Third Class (Steerage). This indicates clear structural privileges, such as closer physical proximity to the boat deck and preferential access to lifeboats.

4.4 Intersectional Analysis: Class, Gender, and Survival Dynamics

Evaluating the intersectional plot generated through `sns.catplot(x='Pclass', y='Survived', col='Initial')` highlights that the protection of female passengers was consistently maintained across the social hierarchy:

- First and Second Class female profiles reached high survival probabilities (95%).
- Third Class female survival dropped significantly to nearly 50%, showing that economic disadvantage degraded the safety protections typically granted to gender.
- For male cohorts, survival remained low across all settings, though First Class males still maintained a higher survival rate (36%) compared to the extreme mortality seen in Third Class males (13%).

he Biological Age Threshold Distribution

Plotting overlapping age histograms via `FacetGrid` reveals crucial survival trends:

- **Infant Target Protection:** Infants and toddlers aged <= 4 exhibited high survival counts, regardless of their family's social class.
- **The Young Adult Attrition Zone:** Individuals between 15 and 35 years old experienced the highest volume of total deaths.
- **Elderly Extremes:** The oldest documented individual on board, an 80-year-old passenger, successfully survived, proving that specific age extremes received priority assistance.

4.6 Kinship Density Analysis (`SibSp` & `Parch`)

Evaluating family size configurations shows that traveling completely alone or within a very large family group noticeably reduced survival chances:

- **The Optimal Kinship Buffer:** Individuals traveling with 1 or 2 family members achieved the highest survival rates (53.58% and 46.42%, respectively).
- **The Steerage Family Trap:** Survival dropped to 0.00% for large extended family units containing >=5 siblings or spouses. These large groups struggled to navigate the ship's internal corridors and secure multiple open spots on a single lifeboat.

Step 4: Feature Engineering & Formatting

1. Title Extraction & Mapping: Extracted passenger titles (e.g., Mr, Mrs, Miss, Master) from the Name feature using regular expressions. To prevent overfitting, `Initial` and high-status titles—including Lady, Countess, Capt, Col, Don, Dr, Major, Rev, Sir, Jonkheer, and Mlle—were consolidated into a single uniform group labeled as Rare. Additionally, titles with identical contextual meanings were standardized (e.g., Mlle and Ms were mapped to Miss, and Mme was mapped to Mrs)

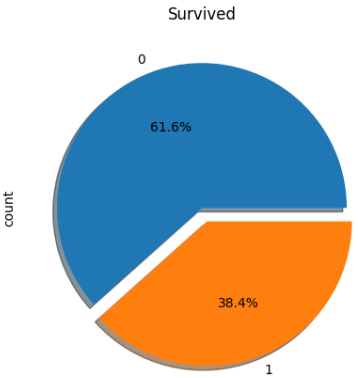
- 7. CONCLUSION AND ANALYTICAL FINDINGS
- This comprehensive Exploratory Data Analysis confirms that survival during the Titanic disaster was not a matter of random chance. Instead, it was deeply shaped by institutional social structures.
- The analytical evidence proves that:
- **Gender and Socio-Economic Class** stood as the two most powerful indicators of survival. Biological gender gave female passengers a clear priority across all classes, but their survival rates were still heavily influenced by economic status.

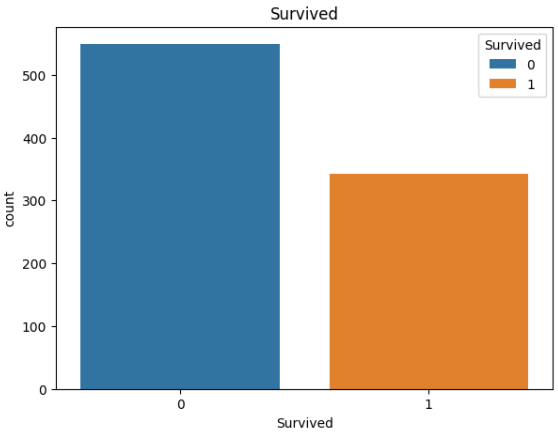
First Class females enjoyed near-total survival, whereas Third Class females were exposed to much higher rates of mortality.

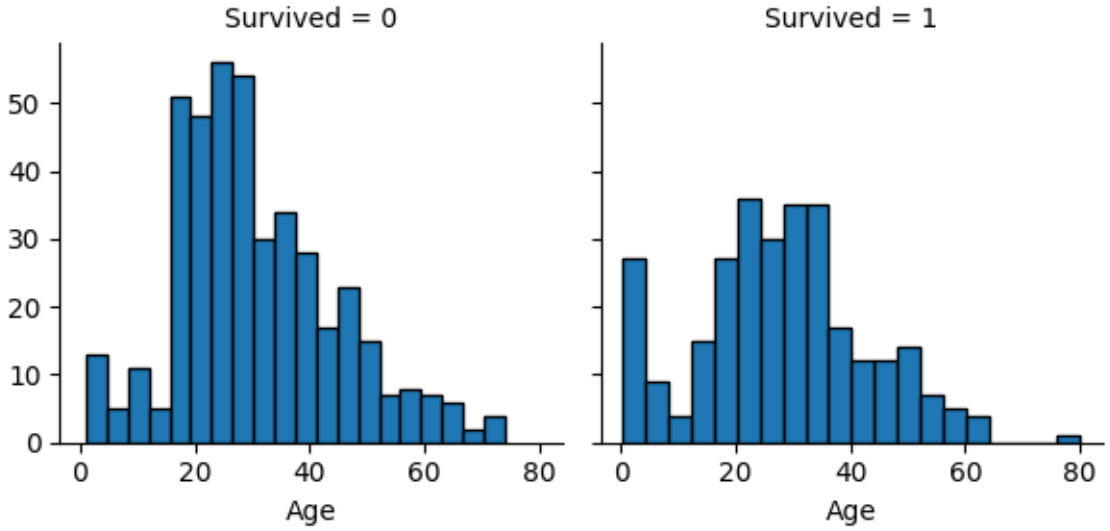
- **Age Dynamics** confirm a clear effort to protect children, as infants and toddlers achieved high survival frequencies across the board.
- **Family Size Configurations** revealed a distinct nonlinear pattern. Moderate family groups (1 to 2 relatives) served as a helpful safety buffer during evacuation logistics. In contrast, large extended family units faced high mortality rates, showing how difficult it was to keep large groups together during a chaotic evacuation.
- Through this data preparation pipeline, the initial fragmented dataset was successfully cleaned of missing entries, stripped of uninformative features, and transformed into a highly structured, fully numeric matrix ready for reliable statistical evaluation

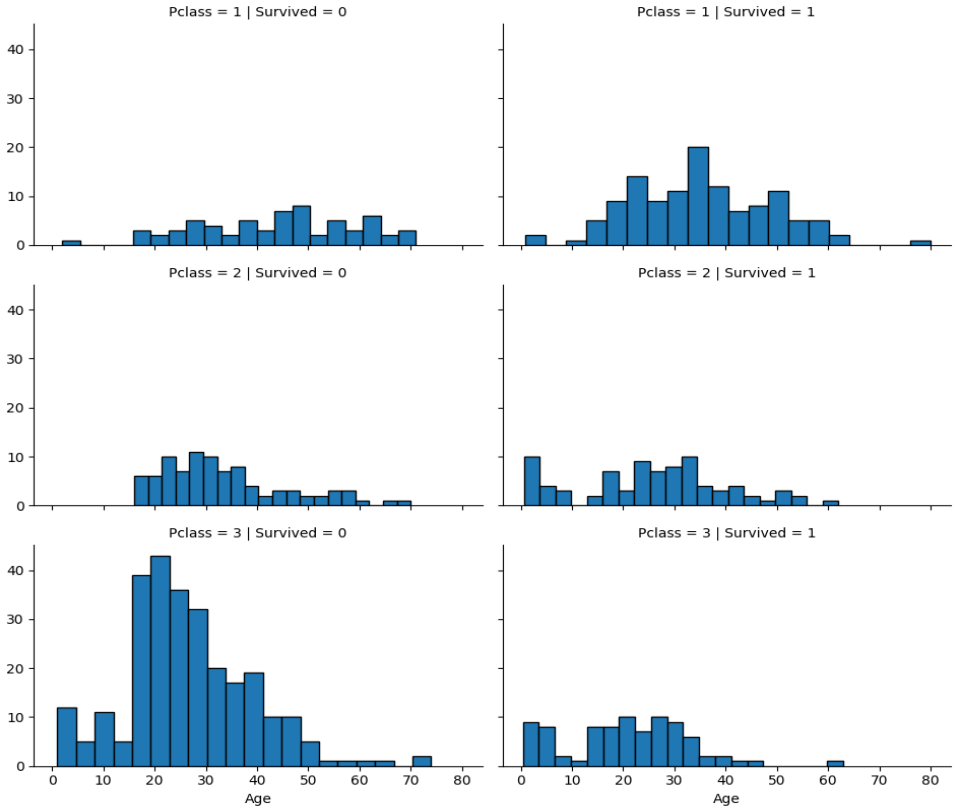
Conclusion:

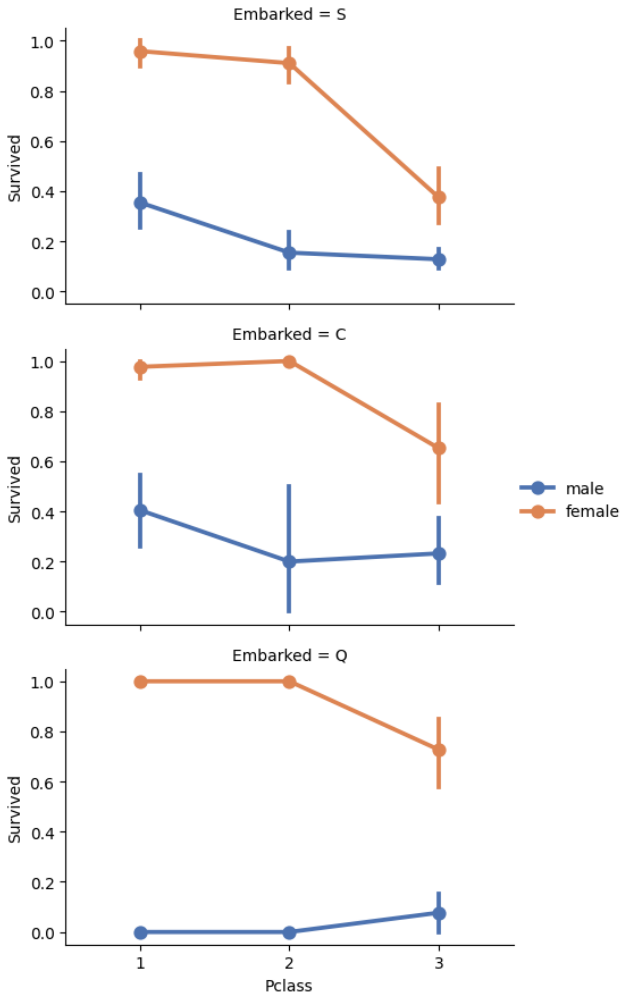
The data proves that survival was not purely random. Socio-economic privilege (Pclass), demographic characteristics (Sex, Age), were heavily intertwined with a passenger's chance of surviving the Titanic disaster.

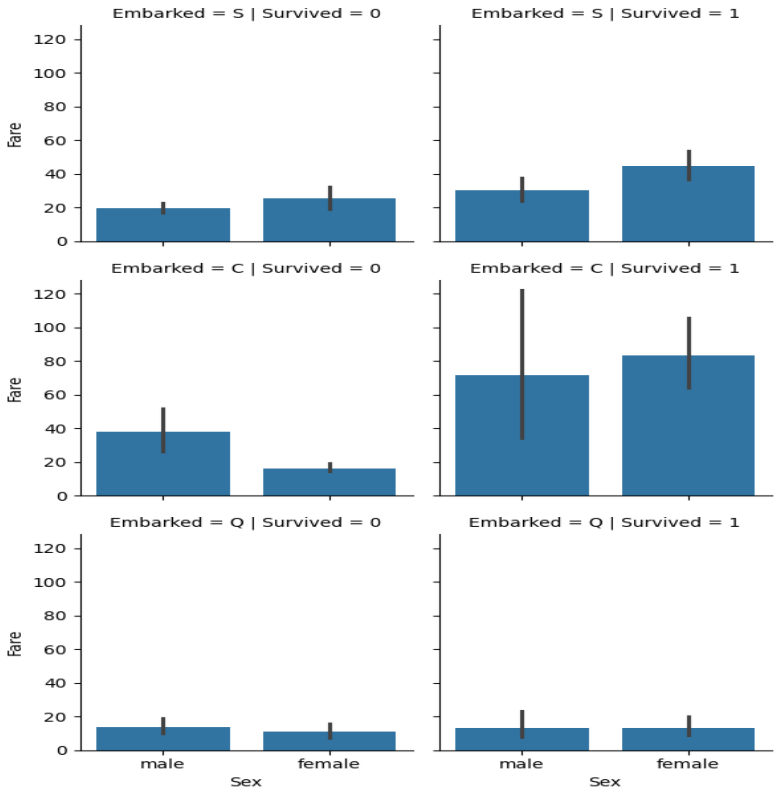

Survived
0
61.6%
38.4%
1
count

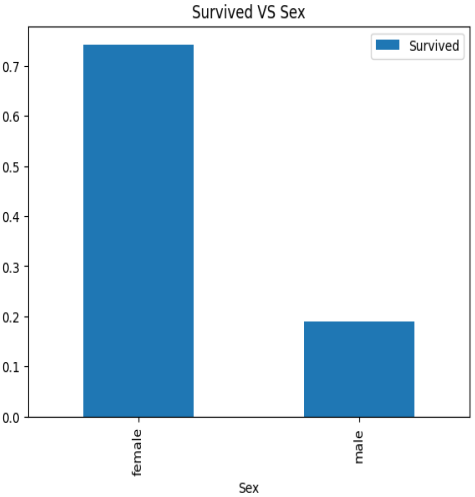

Survived
Survived
0
1
count
500
400
300
200
100
0
0
1
Survived

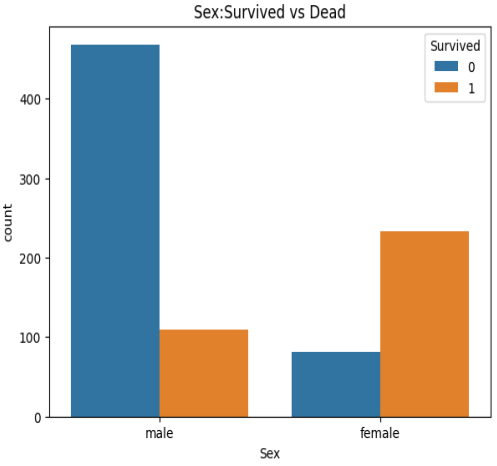

Survived = 0
Survived = 1
50
40
30
20
10
0
0
20
40
60
80
Age
0
20
40
60
80
Age

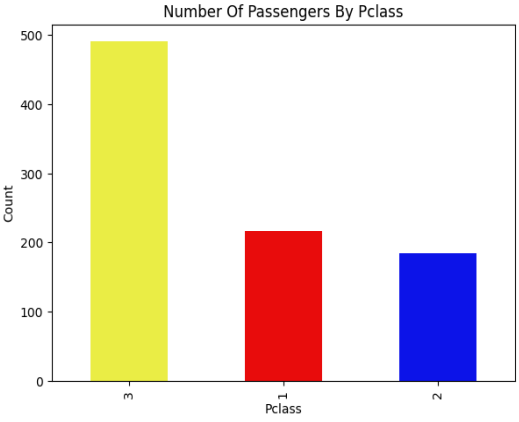

Pclass = 1 | Survived = 0
Pclass = 1 | Survived = 1
Pclass = 2 | Survived = 0
Pclass = 2 | Survived = 1
Pclass = 3 | Survived = 0
Pclass = 3 | Survived = 1
Age
Age

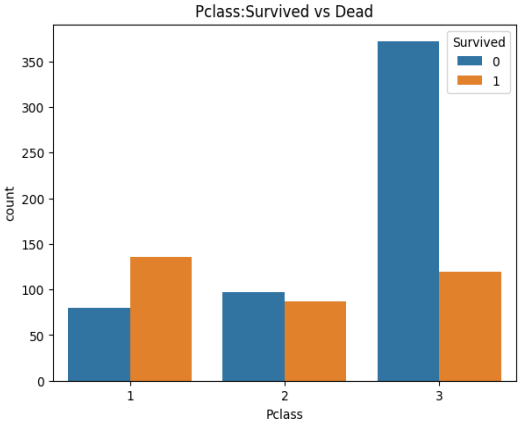

Embarked = S
Embarked = C
Embarked = Q
Survived
Pclass
male
female

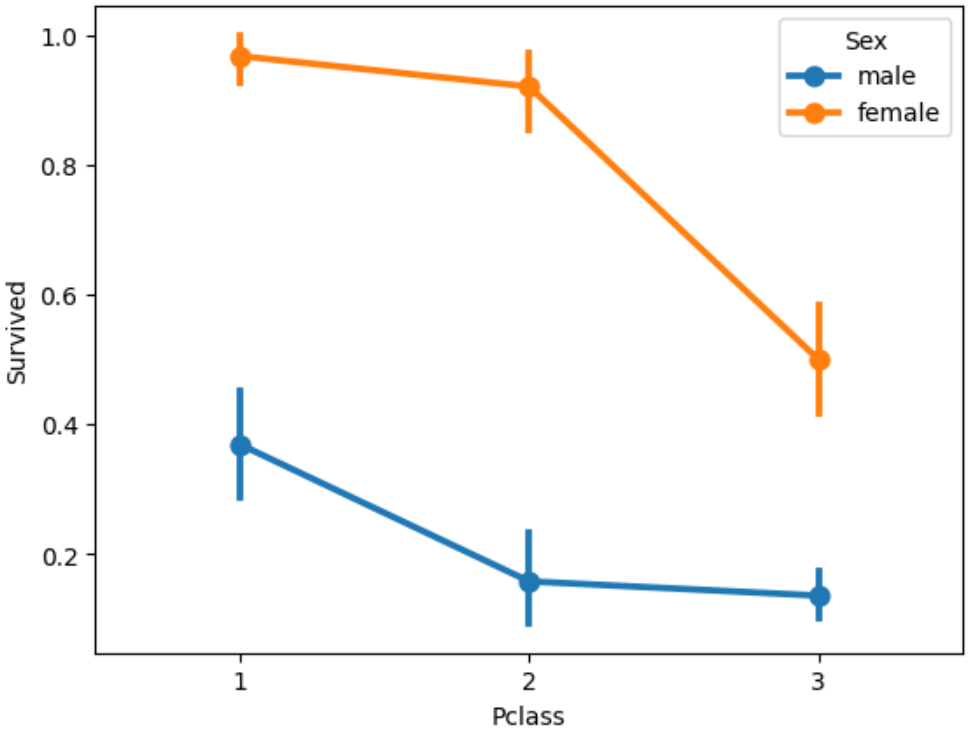

Embarked = S | Survived = 0
Embarked = S | Survived = 1
Embarked = C | Survived = 0
Embarked = C | Survived = 1
Embarked = Q | Survived = 0
Embarked = Q | Survived = 1
Fare
Sex
male
female

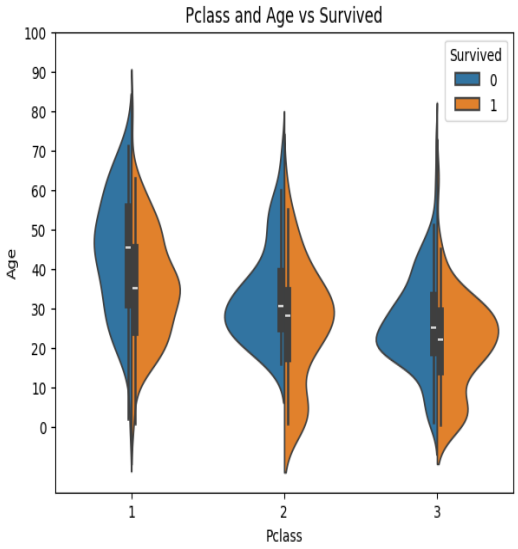

Survived VS Sex
Survived
Sex
female
male

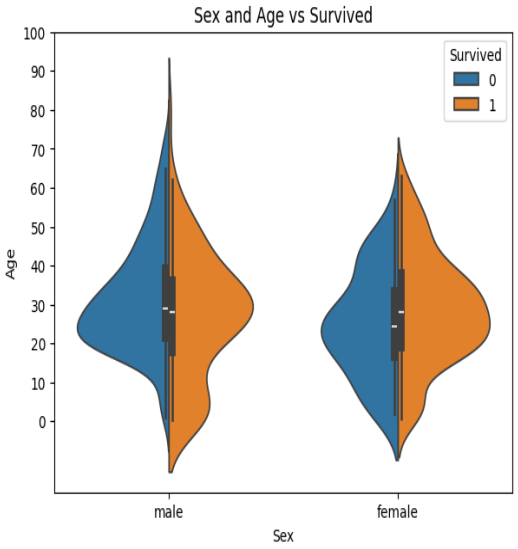

Sex:Survived vs Dead
Survived
0
1
count
Sex
male
female

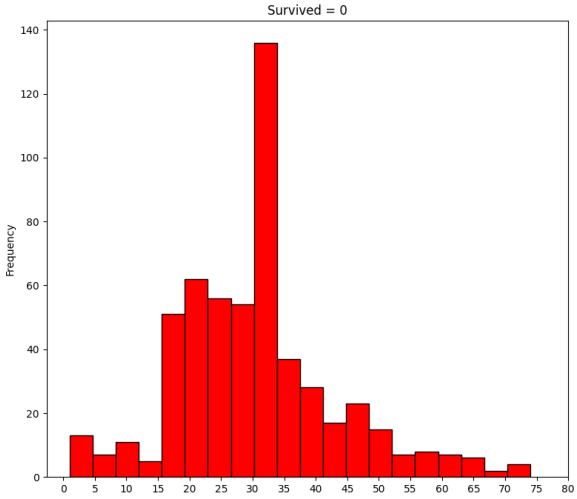

Number Of Passengers By Pclass
Count
Pclass

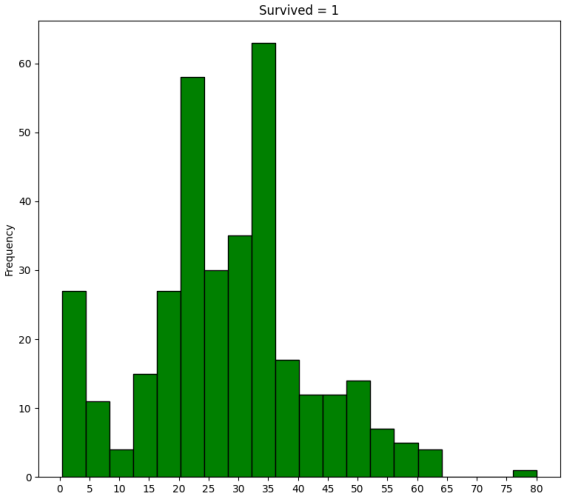

Pclass:Survived vs Dead
Survived
0
1
count
Pclass

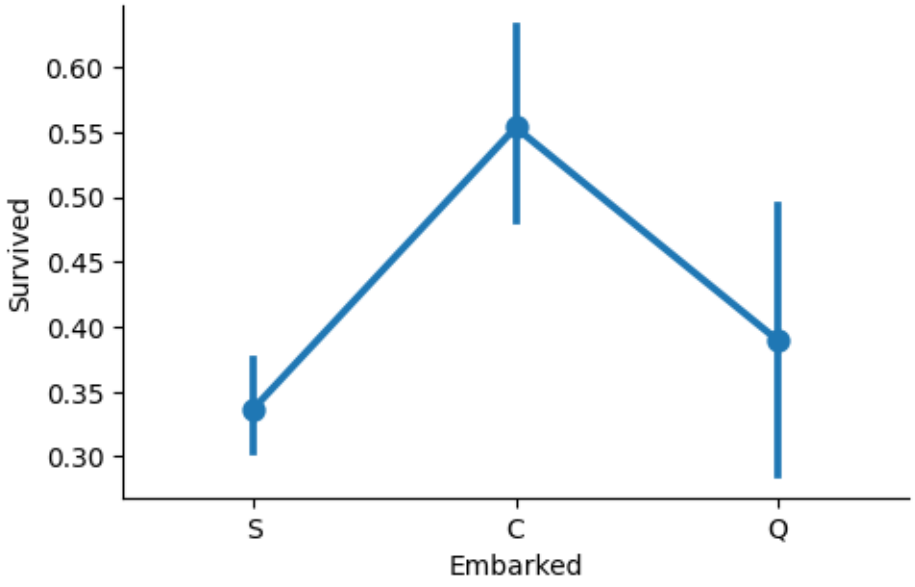

Sex
male
female
Survived
1.0
0.8
0.6
0.4
0.2
1
2
3
Pclass

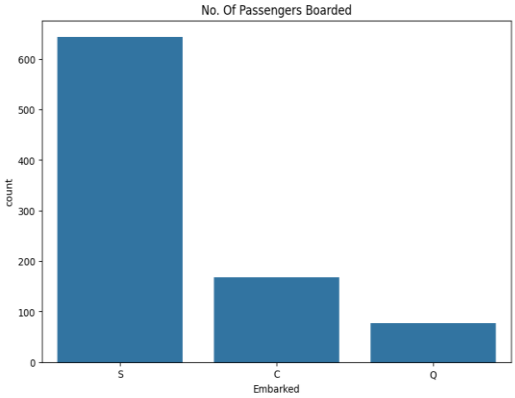

Pclass and Age vs Survived
Survived
0
1
Age
Pclass
1
2
3

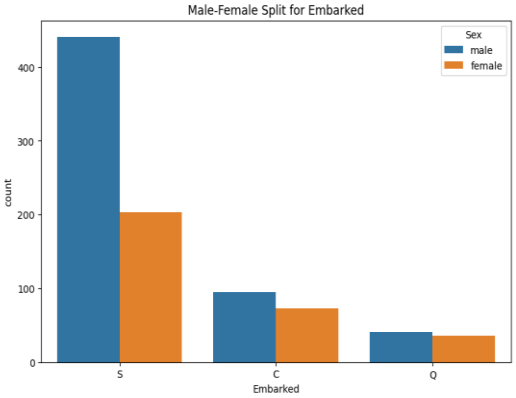

Sex and Age vs Survived
Survived
0
1
Age
Sex
male
female

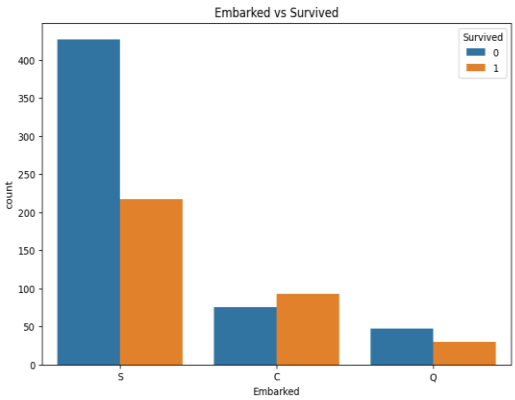

Survived = 0
Frequency
0
20
40
60
80
100
120
140
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80

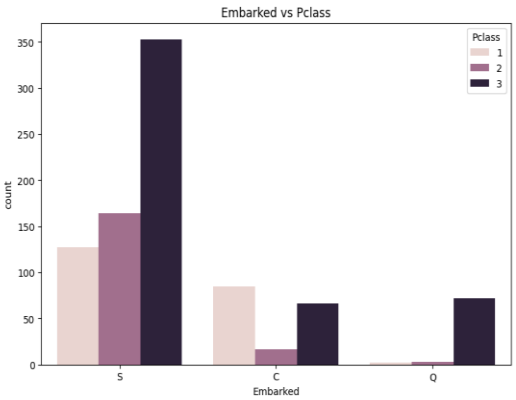

Survived = 1
Frequency
0
10
20
30
40
50
60
0 5 10 15 20 25 30 35 40 45 50 55 60 65 70 75 80


Survived
0.30
0.35
0.40
0.45
0.50
0.55
0.60
S
C
Q
Embarked


No. Of Passengers Boarded
count
Embarked
S
C
Q


Male-Female Split for Embarked
Sex
male
female
count
Embarked
S
C
Q


Embarked vs Survived
Survived
0
1
count
Embarked
S
C
Q


Embarked vs Pclass
Pclass
1
2
3
count
Embarked
S
C
Q